

Assignment 2

Total: 5 marks

This assignment must be hand-written. **Show ALL steps in ALL questions.**
Read the questions carefully. **Submit the graded part only.**

1. **(Graded)** Simplify the following boolean expression: $(A + B)(C+B)(C+A)$ [1 marks]

2. **(Ungraded)** Find the complement of the following expression:
 $(x + y' + z')(x' + y)(x' + z')$

3. **(Graded)** Draw the following functions using NOR gates only (You must use least amount of NOR gates): [2 marks]

$$F(A,B,C,D) = (A'BCD' + A'C' + A(B'+D'))$$

NB: You can't simplify the above function. You have to draw based on the function given in question.

4. **(Ungraded)** Draw the following functions using NAND gates only:

$$F(A,B,C,D) = (A' B' + C'D + CD + (B'C + D'))$$

NB: You can't simplify the above function. You have to draw based on the function given in question.

Find out SOP and POS for the following: (Do not use truth table, Derive POS using Distributive law)

5. **(Graded)** $F(A,C,B) = A'C + BC'$ [1 marks]

6. **(Graded)** $F(A,B,C,D) = A + B'C' + D$ [1 marks]

7. **(Ungraded)** $F(A,B,C,D,E) = ACE + CDE$